

# GFIS Publication-Quality Deliverable Blueprint

## Executive summary

Green Fuel Intelligence System (GFIS) is best positioned as a **physics-guided AI digital twin for anaerobic digestion**, centered on methane-yield prediction, soft sensing of stability indicators, what-if simulation, and constrained operational optimization. That positioning is technically defensible because anaerobic digestion is nonlinear and multistage, ADM1 remains the reference mechanistic model for AD, recent plant-scale studies show that machine learning can forecast biogas from operational and SCADA data, physics-informed machine learning improves plausibility and data efficiency, and digital twins are now widely understood as continuously updated virtual representations used for prediction and decision support. 1

I began with the enabled Google Drive connector and confirmed that your Drive already contains the core BITS/WILP guidance set needed to anchor the entire deliverable stack: the sample abstract/outline report, the WILP report-preparation guidelines, the final-report checklist, the Viva/E-Learn submission steps, and the 2026 dissertation orientation deck. Those documents are sufficiently specific to shape the **GFIS abstract/outline, mid-semester package, final dissertation, and submission logistics**. 2 3 4 5

6

The strongest execution strategy is to build GFIS around a **hybrid data backbone**: one real, open, industrially relevant dataset as the empirical anchor; one synthetic scenario engine to expand coverage safely; one reproducible modeling stack built around **LSTM + XGBoost + physics constraints**; one **VFA/ALK soft sensor** producing both regression and stability classes; and one lightweight deployment layer using **FastAPI + Streamlit + SQLite**. The University of Iowa Muscatine plant dataset is the best currently visible open time-series anchor because it contains cleaned laboratory, operational, and minute-resolution SCADA data from a full-scale co-digestion facility across 2020–2023. 7

One practical limitation matters immediately: the Google Drive connector exposed in this environment is **read-only** for search, fetch, export, and listing. I can therefore design the complete folder structure, naming specification, and upload sequence precisely, but I cannot programmatically create folders or write files into Drive from here. The storage plan below is therefore written as an exact **manual upload blueprint** that you can execute directly in Drive with no ambiguity.

## Drive audit and BITS submission frame

The BITS/WILP orientation materials in your Drive make the evaluation logic very clear. A dissertation is expected to be **non-trivial, analytical, application-oriented, original, approximately 16 weeks of work, and comparable in effort to four M.Tech courses**. The same orientation deck also states the assessment structure: **Abstract/Outline 10%, Mid-Semester Report 30%, Final Report 60%**, with RRA possible if the work is technically weak, incomplete, or plagiarized. 6

The WILP report guidelines further specify the final-report mechanics: the report is a formal technical document; pages should be double-spaced with about 1-inch margins; the soft copy should be in PDF, below 10 MB, text-selectable rather than scanned wholesale, and original; the formal report should include cover, title page, acknowledgements, abstract sheet, table of contents, introduction, main text, conclusions/recommendations, appendices, references, and glossary. Importantly for your final dissertation, the **abstract sheet should be concise and should not exceed 200 words**, while including project area and keywords. <sup>8</sup>

The E-Learn/Viva instructions in your Drive also clarify the workflow: the student enters the **Viva / Project Work** section, uses **Document Submission**, and uploads the **Abstract, Mid Semester Report**, and **Final Report** through the corresponding menu items. The same instructions note that the final submission stage includes both **Final Report** and **Final Presentation**, along with the **Mentor Evaluation Form**. <sup>5</sup>

The practical implication for GFIS is that you do not need one generic “project file.” You need a **stack of linked artifacts**, each shaped for a specific BITS checkpoint.

| BITS artifact             | What BITS expects                                                                                                                        | GFIS implication                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Abstract / Outline Report | A structured outline with cover details plus background, objectives, scope, plan, literature, supervisor details, and supervisor remarks | Your GFIS outline must stay concise, method-centered, and evaluable as a semester plan rather than as a full thesis |
| Mid-Semester package      | Progress review with report, demo readiness, and presentation                                                                            | By mid-semester, you need working data pipelines, baseline models, initial metrics, and one live dashboard/API demo |
| Final Dissertation Report | Formal technical report with abstract sheet, numbered figures/tables, literature survey, appendices, and references                      | The dissertation must read as a complete research narrative, not as a slide deck or software manual                 |
| Final Presentation        | Technical presentation and defensible demo                                                                                               | The software artifact must be demonstrable, explainable, and linked directly to the research claims                 |

The sample outline document in your Drive is also useful in a subtler way: it frames the **Background** section as the part that explains the relevance of the project to the **current work environment / employing organization**. Even if GFIS is primarily research-driven, that wording is worth preserving in the BITS outline: you should explicitly connect GFIS to sustainable energy analytics, digital monitoring, and AI-enabled operational decision support for process systems.

## Drive folder structure and deliverable map

The recommended Drive root should be:

GFIS\_2026\_Akash\_Chatake/

- |— 00\_ADMIN/
  - | |— GFIS\_Project\_Charter\_v01.docx
  - | |— GFIS\_Guide\_Meeting\_Notes\_v01.docx
  - | |— GFIS\_Deliverables\_Register\_v01.xlsx
  - | |— GFIS\_Reference\_Master\_Bibliography\_v01.bib
  - | |— GFIS\_Reference\_Master\_Bibliography\_v01.ris
  - | |— GFIS\_Artifact\_Metadata\_v01.csv
- |— 01\_BITS\_ABSTRACT\_OUTLINE/
  - | |— GFIS\_Abstract\_Outline\_BITS\_v01.docx
  - | |— GFIS\_Abstract\_Outline\_BITS\_v01.pdf
  - | |— GFIS\_Abstract\_Explainer\_Guide\_v01.docx
  - | |— GFIS\_Abstract\_Explainer\_Guide\_v01.pdf
  - | |— GFIS\_Outline\_Presentation\_v01.pptx
- |— 02\_MID\_SEM/
  - | |— GFIS\_MidSem\_Report\_v01.docx
  - | |— GFIS\_MidSem\_Report\_v01.pdf
  - | |— GFIS\_MidSem\_Presentation\_v01.pptx
  - | |— GFIS\_MidSem\_Demo\_Script\_v01.md
  - | |— GFIS\_MidSem\_Results\_Snapshot\_v01.xlsx
- |— 03 DISSERTATION/
  - | |— GFIS\_Dissertation\_Book\_v01.docx
  - | |— GFIS\_Dissertation\_Book\_v01.pdf
  - | |— GFIS\_Dissertation\_Figures/
  - | |— GFIS\_Dissertation\_Tables/
  - | |— GFIS\_Appendices/
  - | |— GFIS\_Checklist\_WILP\_Final\_v01.pdf
- |— 04\_JOURNAL\_MANUSCRIPT/
  - | |— GFIS\_Journal\_Manuscript\_v01.docx
  - | |— GFIS\_Journal\_Manuscript\_v01.pdf
  - | |— GFIS\_Journal\_Cover\_Letter\_v01.docx
  - | |— GFIS\_Journal\_Response\_To\_Reviewers\_Template\_v01.docx
  - | |— GFIS\_Journal\_Figures/
  - | |— GFIS\_Journal\_Tables/
  - | |— GFIS\_Journal\_Supplementary\_Material\_v01.pdf
- |— 05\_DATA/
  - | |— raw/
    - | | |— muscatine\_scada\_raw/
    - | | |— muscatine\_lab\_raw/
    - | | |— epa\_facility\_context/
    - | | |— bmp\_reference\_sources/
  - | |— interim/
  - | |— processed/
  - | |— synthetic/
  - | |— schemas/

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|   |   | data_dictionary/
|   |   | └─ GFIS_Data_ReadMe_v01.md
|   └─ 06_CODE/
|       |   |   | src/
|       |   |   | └─ gfis/
|       |   |   |     |   | data/
|       |   |   |     |   | features/
|       |   |   |     |   | models/
|       |   |   |     |   | physics/
|       |   |   |     |   | soft_sensor/
|       |   |   |     |   | twin/
|       |   |   |     |   | optimization/
|       |   |   |     |   | explainability/
|       |   |   |     |   | api/
|       |   |   |     |   └─ ui/
|       |   | notebooks/
|       |   | tests/
|       |   | configs/
|       |   | scripts/
|       |   | Dockerfile
|       |   | docker-compose.yml
|       |   | requirements.txt
|       |   | environment.yml
|       |   | pyproject.toml
|       |   └─ README.md
|   └─ 07_MODELS/
|       |   | baselines/
|       |   | xgboost/
|       |   | lstm/
|       |   | ensemble/
|       |   | calibrators/
|       |   | model_cards/
|       |   └─ registry.csv
|   └─ 08_RESULTS/
|       |   | benchmark_metrics/
|       |   | ablation_studies/
|       |   | robustness_tests/
|       |   | statistical_tests/
|       |   | shap_outputs/
|       |   | calibration/
|       |   └─ optimization_runs/
|   └─ 09_DEPLOYMENT/
|       |   | fastapi_service/
|       |   | streamlit_dashboard/
|       |   | react_frontend_optional/
|       |   └─ sqlite_db/

```

```

|   |— deployment_docs/
|   |— screenshots/
|— 10_REPRODUCIBILITY/
|   |— seeds/
|   |— split_manifests/
|   |— experiment_registry/
|   |— hardware_software_manifest/
|   |— LICENSE
|   |— CITATION.cff
|   |— Reproducibility_Checklist_v01.md

```

The structure above is intentionally dissertation-first, paper-ready, and audit-friendly. It separates **narrative outputs, data provenance, code, trained artifacts, evaluation evidence**, and **deployment**—which is exactly what reviewers, faculty, and later journal editors will ask for.

The deliverable map below turns that folder tree into an execution matrix.

| Deliverable                | Mandatory sections/<br>components                                                                                                     | Figures/tables<br>required                                  | Code/data<br>dependencies                     | Acceptance<br>standard                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------|
| BITS Abstract /<br>Outline | Cover, background,<br>objectives, scope, plan<br>of work, literature,<br>particulars of<br>supervisor/additional<br>examiner, remarks | 1 system block<br>diagram, 1 plan<br>table                  | None beyond<br>conceptual<br>architecture     | Must look<br>feasible,<br>original, and<br>guide-<br>discussion<br>ready |
| Abstract<br>Explainer Doc  | Problem statement,<br>why hybrid AI, why<br>physics-guided, why<br>digital twin, expected<br>outputs                                  | 1 concept<br>flowchart, 1<br>example scenario<br>table      | None                                          | Must help you<br>defend the<br>outline orally                            |
| Mid-Sem<br>Report          | Literature, data<br>pipeline, baseline<br>experiments, early<br>architecture, interim<br>results, next-step plan                      | Data schema,<br>train-test split,<br>baseline<br>comparison | Real data<br>ingest,<br>preliminary<br>models | Must prove<br>tangible<br>progress and<br>demo<br>readiness              |
| Dissertation /<br>Book     | Full theory, literature<br>review, methodology,<br>experiments,<br>discussion,<br>deployment,<br>conclusion, appendices               | Full figure pack,<br>full table pack                        | Entire codebase<br>and results                | Submission-<br>ready technical<br>report                                 |

| Deliverable          | Mandatory sections/<br>components                                            | Figures/tables<br>required        | Code/data<br>dependencies                 | Acceptance<br>standard            |
|----------------------|------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------|-----------------------------------|
| Journal manuscript   | IMRaD + methods + results + discussion + limitations + supplementary methods | Selected high-impact figures only | Curated experiments and manuscript tables | Q2/Q3 submission ready            |
| Codebase             | Modular package, configs, tests, readme, API, UI                             | Architecture diagram only         | Full data and experiment registry         | Re-runnable by another researcher |
| Datasets             | Raw/interim/processed/synthetic, schema, data dictionary                     | Dataset summary tables            | Download and processing scripts           | Traceable and versioned           |
| Deployment bundle    | FastAPI, dashboard, SQLite, Docker, screenshots                              | Deployment architecture figure    | Final models and processed data           | Demo-ready and explainable        |
| Reproducibility pack | splits, seeds, configs, manifest, environment                                | Reproducibility table             | Complete experiment metadata              | Publication support artifact      |

## Dissertation and manuscript plan

### Dissertation/book outline

A strong GFIS dissertation should read like a compact research monograph rather than a conventional software project report. The chapter plan below keeps the work technically deep but still achievable within the BITS semester frame.

| Chapter   | Working title                        | Core content                                                                                                                                                             |
|-----------|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chapter 1 | Introduction and research motivation | Renewable energy context; AD complexity; need for forecasts, stability monitoring, and actionable control; problem gap between ADM1 realism and black-box ML convenience |
| Chapter 2 | Fundamentals of anaerobic digestion  | Hydrolysis, acidogenesis, acetogenesis, methanogenesis; operational variables; methane as performance target; process instability indicators                             |
| Chapter 3 | Literature review                    | ADM1 and mechanistic models; ML for biogas prediction; digital twin concepts; soft sensing in AD; physics-informed ML; explainability for environmental AI               |

| Chapter    | Working title                          | Core content                                                                                                                               |
|------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Chapter 4  | Research problem and hypothesis        | Research questions; formal problem definition; target variables; assumptions; novelty claims; deliverable hypotheses                       |
| Chapter 5  | Data design and engineering            | Real dataset selection; schema design; synthetic data generation; preprocessing; feature engineering; leakage control; target construction |
| Chapter 6  | GFIS methodology                       | LSTM sequence model; XGBoost tabular model; ensemble strategy; physics bounds; soft-sensor module; digital twin logic; optimization module |
| Chapter 7  | Experimental protocol                  | Time-aware splits; baseline models; training protocol; evaluation metrics; calibration; robustness analysis; statistical tests; ablations  |
| Chapter 8  | Results and discussion                 | Regression accuracy; physics consistency; soft-sensor performance; explainability; what-if scenarios; optimization outcomes; failure cases |
| Chapter 9  | Software implementation and deployment | FastAPI services; dashboard; SQLite store; model registry; API contracts; demo workflows                                                   |
| Chapter 10 | Conclusions, limitations, future work  | Key findings; practical contribution; limits of synthetic data; roadmap to real plant deployment and IoT integration                       |
| Appendices | Supporting material                    | BITS abstract sheet, data dictionary, hyperparameter tables, extra plots, API schema, ethics/plagiarism declaration, checklist             |

## Journal manuscript outline and target journals

The manuscript should not be “the dissertation cut down.” It should be a purpose-built paper with one crisp claim:

**A physics-guided hybrid digital twin can improve methane-yield forecasting while reducing physically implausible predictions and supporting stability-aware scenario analysis in anaerobic digestion.**

A Scopus-oriented journal plan that balances fit and acceptance probability is below. Quartiles and scopes should always be rechecked at the submission date because journal rankings can shift by category and year.

| Journal                                           | Current positioning                                                                                                                                                       | Why it fits GFIS                                                                                 | Submission strategy                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Bioresource Technology Reports</b>             | SCImago 2024 Q2; scope emphasizes bioresource management, sustainable biotechnology, bioengineering, circular bioeconomy, and bio-based energy/fuels. <sup>9</sup>        | Best primary target for an applied AD + ML + deployment paper                                    | Lead with application value, reproducibility, and full-scale dataset anchoring        |
| <b>Biofuels, Bioproducts and Biorefining</b>      | SCImago 2024 Q2; scope centers on sustainable products, fuels, energy, and systems-based bio-renewables research. <sup>10</sup>                                           | Good fit if the paper emphasizes sustainability systems, bioenergy, and operational implications | Frame the work as an AI-enabled decision-support system for biomethane optimization   |
| <b>Biotechnology for Biofuels and Bioproducts</b> | SCImago shows Q2 in Renewable Energy, Sustainability and the Environment for 2024, and the official scope explicitly includes anaerobic digestion of waste. <sup>11</sup> | Strong thematic fit, but it can behave like a stretch target depending on category               | Submit if experiments are especially strong and the AD contribution is clear          |
| <b>Discover Applied Sciences</b>                  | SCImago 2024 Q2, multidisciplinary applied-science venue. <sup>12</sup>                                                                                                   | Backup venue if you want a broader applied AI + sustainability framing                           | Emphasize digital twin architecture, explainability, and deployment artifactual value |

The paper itself should follow this structure:

| Section               | What must be included                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------|
| Title                 | Physics-guided hybrid AI digital twin; methane prediction; anaerobic digestion                        |
| Abstract              | Problem, method, dataset, best metrics, physics consistency gain, decision-support takeaway           |
| Introduction          | AD challenge, literature gap, contributions                                                           |
| Related work          | ADM1, plant-scale ML, soft sensing, digital twins, physics-informed ML                                |
| Materials and methods | Datasets, preprocessing, features, LSTM/XGBoost/ensemble, physics constraint, soft sensor, evaluation |
| Results               | Forecasting, classification, calibration, ablations, robustness, scenario analysis                    |
| Discussion            | Why the hybrid worked; when it fails; practical relevance; comparison with literature                 |

| Section                | What must be included                                                |
|------------------------|----------------------------------------------------------------------|
| Conclusion             | Scientific and applied contribution, limitations, future work        |
| Supplementary material | Hyperparameters, data schema, extra figures, reproducibility details |

## Experiments, metrics, ablations, and reproducibility for the paper

| Experimental package        | Required content                                                                       |
|-----------------------------|----------------------------------------------------------------------------------------|
| Forecasting baselines       | Mean predictor, linear regression, ridge/lasso, random forest, pure XGBoost, pure LSTM |
| GFIS core models            | XGBoost only, LSTM only, weighted ensemble, stacked ensemble, physics-guided ensemble  |
| Soft sensor study           | VFA/ALK regression; thresholded stability classification; multi-task variant optional  |
| Physics study               | Unconstrained vs post-hoc clipping vs physics penalty vs both                          |
| Robustness study            | Reduced training data, missing-value stress, sensor noise, moderate distribution shift |
| Horizon study               | Same-step, short horizon, and multi-step-ahead methane forecasts                       |
| Explainability study        | Global SHAP ranking; local case explanations for stable vs critical states             |
| Scenario/optimization study | What-if simulation for OLR, temperature, HRT, moisture, and pH regime changes          |

The evaluation metrics should be fixed in advance:

| Task                     | Metrics                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------|
| Methane regression       | MAE, RMSE, $R^2$ , sMAPE or MAPE only if denominator stability is acceptable                                    |
| Physical consistency     | Violation rate, mean excess over bound, negative prediction rate, clipped-mass fraction                         |
| VFA/ALK regression       | MAE, RMSE, $R^2$ , interval coverage if uncertainty is modeled                                                  |
| Stability classification | Precision, Recall, F1 (macro and weighted), confusion matrix, AUROC or one-vs-rest ROC where meaningful         |
| Probability calibration  | Expected Calibration Error and reliability diagrams; temperature scaling can be tested if needed. <sup>13</sup> |

The statistical-comparison plan should include **rolling-origin evaluation**, **bootstrap 95% confidence intervals** for headline metrics, **Wilcoxon signed-rank tests** on paired forecast errors, and **Diebold-Mariano tests** if you compare forecast-loss sequences across horizons. For the multiclass stability classifier, use bootstrap confidence intervals for macro-F1 and, where useful, test paired error differences rather than relying only on a single confusion matrix.

The reproducibility checklist should be explicit and non-negotiable:

1. Fixed train/validation/test split manifest
2. Random seed registry
3. Locked environment file
4. Full config files for each experiment
5. Model cards for final models
6. Data dictionary and schema
7. Git commit hash recorded in all result folders
8. CPU/GPU and runtime log
9. Hyperparameter report
10. Exact figure-generation scripts

## Methodology and system architecture

Anaerobic digestion is well suited to a hybrid design because the process is governed by science that is partially known and partially hidden in plant behavior. ADM1 remains the canonical process model: it formalizes the biochemical and physicochemical conversions in AD and, in one common DAE implementation, contains **26 dynamic state variables and 8 algebraic variables**. That interpretability is valuable, but the model's dimensionality and calibration burden make it awkward for fast operational decision support. In contrast, ML models can forecast plant behavior from observed histories, as shown in recent full-scale SCADA work, but unconstrained ML can drift into physically implausible domains. <sup>14</sup>

GFIS therefore uses a **hybrid surrogate-twin architecture** instead of attempting a full online ADM1 solver. The key design idea is simple: let the AI learn plant-specific temporal and nonlinear relationships, but let physics define feasibility boundaries, state semantics, and scenario constraints. That is fully aligned with the broader physics-informed ML literature, which treats domain laws, invariances, or governing relationships as structural priors that improve plausibility and generalization. <sup>15</sup>

flowchart TD

A[Raw data: SCADA, lab data, plant logs, optional EPA context] --> B[Data QA and preprocessing]

B --> C[Feature engineering]

C --> C1[Tabular features]

C --> C2[Sequence windows]

C --> C3[Scenario variables]

C1 --> D1[XGBoost methane model]

C2 --> D2[LSTM methane model]

C3 --> E1[VFA/ALK soft-sensor regressor]

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E1 --> E2[Stability classifier: Stable / Warning / Critical]

D1 --> F[Ensemble fusion]
D2 --> F
F --> G[Physics-guided constraint layer]
G --> H[Methane forecast]

H --> I[Digital twin state update]
E2 --> I
C3 --> I

I --> J[What-if simulation engine]
J --> K[Optimization module]
H --> L[Explainability layer: SHAP + local cases]
K --> M[Recommended operating region]
L --> N[Dashboard / API]
M --> N
H --> N

```

## Data acquisition and generation

The most defensible real-data option is the **Muscatine Water Resource Recovery Facility** dataset released by the University of Iowa. It contains laboratory, operational, and minute-resolution SCADA data for anaerobic co-digestion from 2020–2023 and is described by the deposit as the largest full-scale facility dataset on biogas production available to the authors. The EPA Anaerobic Digestion Data Collection Project is not a time-series model-training source in the same sense, but it is useful as a contextual dataset for plant typology, feedstocks, processing capacity, and facility attributes. <sup>16</sup>

Because no single public dataset is likely to cover all failure regimes, feedstock chemistries, and stability transitions required for a publication-grade digital-twin study, GFIS should use a **hybrid data strategy**:

| Layer                   | Recommended source                                | Role in GFIS                                                        |
|-------------------------|---------------------------------------------------|---------------------------------------------------------------------|
| Real time-series anchor | Muscatine lab + SCADA data                        | Main empirical training/evaluation backbone                         |
| Facility context        | EPA AD facility survey                            | Narrative context and possible external validation descriptors      |
| Feedstock priors        | BMP reviews and open feedstock/digestion datasets | Priors for methane-potential bounds and substrate class assumptions |
| Synthetic scenarios     | Physics-bounded simulation and regime synthesis   | Coverage expansion for rare instability events and what-if analysis |

A good synthetic-data engine should not merely sample random rows. It should synthesize **coherent plant trajectories**. The recommended procedure is:

1. Define variable ranges for temperature, pH, OLR, HRT, TS, VS, COD, moisture, C/N, alkalinity, gas composition, and ambient conditions.
2. Sample operating regimes using Latin hypercube or stratified scenario grids.
3. Inject temporal correlation through autoregressive state updates and rolling windows.
4. Define stable, warning, and critical regimes through VFA/ALK trajectory rules and physically plausible methane-response curves.
5. Enforce methane upper bounds from theoretical or empirical methane potential.
6. Add realistic missingness, noise, drift, and outlier patterns.
7. Validate synthetic distributions against the real anchor set using marginal comparisons and pairwise dependence checks.

**Preprocessing and feature engineering**

The preprocessing pipeline should be deterministic and versioned:

- time alignment and resampling
- duplicate removal
- sensor-range filtering
- missing-value imputation with masks retained as features
- leakage-safe training windows
- outlier tagging rather than blind deletion
- regime annotation for maintenance/upset periods

Feature engineering is likely to matter as much as the model. In the Muscatine study, feature engineering was essential and most of the most important SCADA features were engineered rather than raw. That finding strongly supports making feature design a first-class component of GFIS rather than an afterthought. <sup>17</sup>

Recommended features include:

| Feature family      | Examples                                                          |
|---------------------|-------------------------------------------------------------------|
| Raw operational     | Temperature, pH, flow, gas rate, TS, VS, COD, alkalinity          |
| Feedstock quality   | C/N, moisture, substrate category, volatile solids fraction       |
| Lagged features     | 1-step, 6-step, 24-step lags, rolling means, rolling slopes       |
| Interaction terms   | OLR × temperature, VS × HRT, pH × alkalinity                      |
| Stability proxies   | VFA trend, alkalinity trend, gas-composition trend, pH volatility |
| Temporal indicators | hour, day, week, season, maintenance flag                         |

**Model architecture**

The core method stack uses three complementary components:

| Module           | Design                                                                    |
|------------------|---------------------------------------------------------------------------|
| Sequence learner | LSTM on sliding windows for temporal dynamics and delayed process effects |
| Tabular learner  | XGBoost on engineered features for nonlinear structured relationships     |
| Fusion           | Weighted late fusion first; stacked meta-learner as an advanced variant   |

LSTM is a natural sequence model because it was designed to preserve long-range dependencies through gated memory cells. XGBoost is a strong tabular benchmark because of its scalable boosting framework, sparsity-aware learning, and well-established performance on structured data. SHAP is the preferred explanation layer because it provides additive feature attributions with a consistent theoretical basis. <sup>18</sup>

A practical methane-forecast stack is:

- **LSTM branch:** input shape `[batch, window, features]`; output one-step or short-horizon methane prediction
- **XGBoost branch:** engineered tabular features from the same timestamp/window summary
- **Fusion:**

$$\hat{y}_t = w_1 \hat{y}_t^{(xgb)} + w_2 \hat{y}_t^{(lstm)}, \quad w_1 + w_2 = 1$$

with weights learned on validation data or by a meta-learner

## Physics-constrained methane formulation

The physics layer should encode one principle that is hard to dispute: **predicted methane cannot exceed feasible methane production implied by substrate availability and methane potential.**

If elemental composition is available, a Buswell-type theoretical methane potential can be used. For a substrate with empirical formula  $C_a H_b O_c N_d$ , a common theoretical methane expression is: <sup>19</sup>

$$TBMP_{CH_4} = \frac{22.4 \left( \frac{a}{2} + \frac{b}{8} - \frac{c}{4} - \frac{3d}{8} \right)}{12a + b + 16c + 14d}$$

where the result is expressed at standard conditions per unit substrate mass after appropriate conversion.

In plant operation, elemental composition is often unavailable at high frequency. GFIS should therefore use a more deployable upper-bound formulation:

$$y_t^{\max} = \eta_t \cdot VS_t \cdot BMP_t^{\text{prior}}$$

where:

- $VS_t$  = volatile solids loading or available volatile solids basis
- $BMP_t^{\text{prior}}$  = feedstock-class methane-potential prior
- $\eta_t \in (0, 1]$  = conversion-efficiency factor reflecting realistic digestion performance, HRT, and regime

The model output is then constrained via either hard clipping or a train-time penalty:

$$\mathcal{L}_{total} = \mathcal{L}_{pred} + \lambda \mathbb{E} \left[ \max(0, \hat{y}_t - y_t^{\max})^2 + \max(0, -\hat{y}_t)^2 \right]$$

A minimal implementation is:

```
import torch

def physics_penalty(y_pred, y_max):
    upper = torch.relu(y_pred - y_max) ** 2
    lower = torch.relu(-y_pred) ** 2
    return (upper + lower).mean()

def total_loss(y_pred, y_true, y_max, lam=0.2):
    pred_loss = torch.mean((y_pred - y_true) ** 2)
    phys_loss = physics_penalty(y_pred, y_max)
    return pred_loss + lam * phys_loss
```

## Soft sensor for VFA/ALK and stability state

The soft sensor is scientifically important because VFA/ALK acts as an operational stability indicator. Literature repeatedly treats VFA-to-alkalinity ratio as a first-line warning metric, with stable operation commonly associated with values below roughly **0.3–0.4**, and substantial instability or failure risk above about **0.8**. <sup>20</sup>

GFIS should encode the stability label policy as:

- **Stable:** VFA/ALK < 0.4
- **Warning:**  $0.4 \leq \text{VFA/ALK} \leq 0.8$
- **Critical:** VFA/ALK > 0.8

The best engineering design is a two-stage soft sensor:

1. **Regression head** to estimate VFA/ALK from easy-to-measure operational variables
2. **Classification head** to map the estimated ratio, and optionally other features, into Stability / Warning / Critical states

This can be implemented either as separate models or a multi-task network with shared representation.

## Digital twin simulation approach

NASA's framing of digital twins as data-linked virtual replicas used to simulate, test, and support decisions is highly compatible with a methane-digestion setting. In GFIS, the digital twin should therefore be treated as an **operational surrogate twin** rather than as a full mechanistic clone. <sup>21</sup>

The simulation loop at each decision horizon should be:

1. Ingest latest plant state and recent window

2. Estimate hidden stability variables by soft sensing
3. Forecast methane and risk over the chosen horizon
4. Apply physics consistency layer
5. Allow what-if perturbations in OLR, temperature, pH control, HRT, moisture, and feed mix
6. Score candidate actions by methane gain vs stability risk
7. Surface the safest high-yield operating band

## Optimization module

The optimization objective should remain modest and defensible:

$$\max_{u \in \mathcal{U}} \hat{CH}_4(u) - \alpha \cdot Risk(u) - \beta \cdot Viol(u)$$

where  $u$  is a candidate operating configuration,  $Risk(u)$  is the predicted instability probability or expected VFA/ALK exceedance, and  $Viol(u)$  is any physics-rule violation term.

A practical implementation can use:

- constrained grid search for the dissertation baseline
- Bayesian optimization as an advanced optional module
- hard safety constraints, for example:
  - pH range
  - temperature regime
  - stability class must not be Critical
  - VFA/ALK predicted threshold must remain below a chosen cap

## Explainability and deployment

SHAP should be used in two layers:

- **global explanations** to show dominant drivers of methane yield and instability
- **local explanations** for critical operational episodes, so the guide and evaluator can see why the model raised a warning or recommended a change. <sup>22</sup>

For deployment, the tool choice is sound and implementation-friendly:

- **FastAPI** for serving prediction and scenario endpoints; it generates OpenAPI-based interactive API docs and supports production-style API organization. <sup>23</sup>
- **Streamlit** for a fast, Python-native dashboard and research-demo interface. <sup>24</sup>
- **SQLite** for a lightweight, serverless, single-file relational store for experiment logs, prediction history, and scenario outputs. <sup>25</sup>

A minimal API skeleton is:

```
from fastapi import FastAPI
from pydantic import BaseModel
import joblib
```

```

app = FastAPI(title="GFIS API", version="1.0.0")

class PredictRequest(BaseModel):
    features: dict
    scenario: dict | None = None

@app.post("/predict/methane")
def predict_methane(req: PredictRequest):
    # load models once in production
    model = joblib.load("models/ensemble/gfis_ensemble.pkl")
    y_pred = model.predict([req.features])[0]
    return {"methane_pred": float(y_pred)}

@app.post("/predict/stability")
def predict_stability(req: PredictRequest):
    clf = joblib.load("models/soft_sensor/stability_classifier.pkl")
    label = clf.predict([req.features])[0]
    return {"stability_state": str(label)}

```

The code organization should be kept clean from day one:

```

src/gfis/
├─ data/           # ingestion, validation, schemas
├─ features/       # lag builders, rolling stats, encoders
├─ models/         # xgboost, lstm, fusion, calibration
├─ physics/        # methane bounds, penalties, validators
├─ soft_sensor/    # VFA/ALK regression + stability classification
├─ twin/          # scenario engine and state updates
├─ optimization/   # search and scoring
├─ explainability/ # SHAP and report plots
├─ api/           # FastAPI routes
└─ ui/            # Streamlit pages

```

## Experimental design and timeline

The evaluation protocol must be **time-aware**. Random row-wise train/test splits are not acceptable for a process-forecasting study because they leak future information. Use rolling-origin or blocked chronological splits for all principal results. The model suite should compare classical baselines, machine-learning baselines, and the full physics-guided GFIS stack.

| Study block                     | Models                                                     | Metrics                                                    | Significance / evidence               |
|---------------------------------|------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|
| Methane regression baseline     | Mean, linear, ridge/lasso, random forest                   | MAE, RMSE, $R^2$                                           | Rolling splits + bootstrap CIs        |
| Tabular and sequence comparison | XGBoost vs LSTM                                            | MAE, RMSE, $R^2$                                           | Paired Wilcoxon on absolute errors    |
| Hybrid benefit                  | XGBoost, LSTM, weighted fusion, stacked fusion             | MAE, RMSE, $R^2$                                           | Same test windows, paired comparison  |
| Physics benefit                 | Unconstrained vs clipped vs penalty vs full physics-guided | Violation rate, mean excess over bound, regression metrics | Show accuracy/ plausibility trade-off |
| Soft-sensor regression          | XGBoost, LSTM, multi-task                                  | MAE, RMSE, $R^2$                                           | Bootstrap CIs                         |
| Stability classification        | Logistic, XGBoost classifier, multi-task head              | Precision, Recall, F1 macro, confusion matrix, ECE         | Paired classification error analysis  |
| Calibration                     | Raw classifier vs temperature scaled                       | ECE, reliability diagram, Brier score                      | Calibration report                    |
| Robustness                      | Best model under noise, missingness, reduced data          | Metric degradation                                         | Stress-test table                     |
| Scenario analysis               | Twin predictions under OLR/temp/HRT changes                | Methane uplift, risk change                                | Case studies                          |
| Optimization                    | Grid/Bayesian constrained search                           | Best safe methane recommendation                           | Engineering interpretation            |

The figure program should be planned before experiments begin, because manuscript and dissertation writing become much easier when every experiment is tied to a figure target.

| Figure ID | Figure / diagram          | Purpose                                      | Suggested tool    |
|-----------|---------------------------|----------------------------------------------|-------------------|
| F1        | Overall GFIS architecture | Top-level project explanation                | Mermaid / draw.io |
| F2        | Data pipeline             | Raw-to-processed lineage and leakage control | Mermaid           |
| F3        | Variable schema table     | Data dictionary summary                      | Markdown / Word   |

| Figure ID | Figure / diagram              | Purpose                          | Suggested tool    |
|-----------|-------------------------------|----------------------------------|-------------------|
| F4        | Time-series windowing diagram | LSTM input explanation           | Mermaid           |
| F5        | Correlation heatmap           | Exploratory analysis             | matplotlib        |
| F6        | Baseline vs GFIS metrics      | Main model comparison            | matplotlib        |
| F7        | Physical-violation comparison | Show benefit of physics layer    | matplotlib        |
| F8        | Reliability diagram           | Calibration analysis             | matplotlib        |
| F9        | Confusion matrix              | Stability classifier performance | matplotlib        |
| F10       | SHAP beeswarm / bar plot      | Global explainability            | SHAP + matplotlib |
| F11       | Local SHAP waterfall          | Case explanation                 | SHAP              |
| F12       | Scenario response curves      | Digital twin what-if analysis    | matplotlib        |
| F13       | Optimization frontier         | Yield vs stability tradeoff      | matplotlib        |
| F14       | Deployment architecture       | API + DB + dashboard             | Mermaid           |
| F15       | 16-week Gantt                 | Execution control                | Mermaid           |
| F16       | Folder/reproducibility map    | Artifact audit trail             | Mermaid / draw.io |

The 16-week implementation plan below is scoped to hit both dissertation and paper readiness within one semester.

| Week | Primary objective                       | Concrete outputs                                       |
|------|-----------------------------------------|--------------------------------------------------------|
| 1    | Problem freeze and literature grid      | final topic frame, contribution map, reference library |
| 2    | Drive structure and project scaffolding | folder tree, repo init, metadata manifest              |
| 3    | Data acquisition and schema design      | raw data download, schema docs, data dictionary draft  |
| 4    | Preprocessing pipeline                  | cleaned datasets, leakage-safe split manifests         |
| 5    | Feature engineering                     | lag features, rolling features, scenario variables     |
| 6    | Baseline regression models              | linear/RF/XGB benchmark table                          |
| 7    | LSTM sequence pipeline                  | training-ready sequence windows and first LSTM runs    |
| 8    | Hybrid fusion model                     | XGB + LSTM ensemble metrics                            |

| Week | Primary objective               | Concrete outputs                                             |
|------|---------------------------------|--------------------------------------------------------------|
| 9    | Physics constraint layer        | methane feasibility bounds and penalty study                 |
| 10   | Soft sensor                     | VFA/ALK regressor and stability labels                       |
| 11   | Classification and calibration  | stability classifier, ECE, reliability plots                 |
| 12   | Digital twin scenario engine    | what-if simulator and case runs                              |
| 13   | Optimization and explainability | safe operating-region search, SHAP outputs                   |
| 14   | Deployment                      | FastAPI, Streamlit, SQLite, screenshots                      |
| 15   | Writing integration             | dissertation chapters + journal draft figures/tables         |
| 16   | Final packaging                 | PDF artifacts, checklist, presentation, reproducibility pack |

gantt

```

title GFIS 16-week execution plan
dateFormat YYYY-MM-DD
axisFormat %W

```

section Research framing

```

Topic freeze and contribution map      :a1, 2026-05-11, 7d
Literature matrix and source library   :a2, after a1, 7d

```

section Data engineering

```

Data acquisition and schema design     :b1, 2026-05-25, 7d
Preprocessing and split manifests      :b2, after b1, 7d
Feature engineering                    :b3, after b2, 7d

```

section Modeling

```

Baseline models                       :c1, 2026-06-15, 7d
LSTM pipeline                         :c2, after c1, 7d
Hybrid fusion                         :c3, after c2, 7d
Physics layer                         :c4, after c3, 7d
Soft sensor and classifier             :c5, after c4, 7d

```

section Twin and analysis

```

Calibration and robustness            :d1, 2026-07-20, 7d
Digital twin + what-if engine         :d2, after d1, 7d
Optimization + SHAP                  :d3, after d2, 7d

```

section Packaging

```

Deployment bundle                     :e1, 2026-08-10, 7d
Dissertation and paper writing         :e2, after e1, 7d
Final QA and submission pack          :e3, after e2, 7d

```

## Storage conventions and priority references

Because the Drive connector available here is read-only, the storage plan must be executed manually. The best practice is to create the exact root folder shown above and then upload every artifact under a strict naming convention:

```
GFIS_<ArtifactType>_<ShortDescription>_<YYYYMMDD>_v01.<ext>
```

Examples:

- GFIS\_Abstract\_Outline\_BITS\_20260506\_v01.pdf
- GFIS\_Dissertation\_Book\_20260831\_v01.docx
- GFIS\_Journal\_Manuscript\_20260831\_v01.docx
- GFIS\_ModelCard\_Ensemble\_20260820\_v01.md
- GFIS\_Results\_AblationStudy\_20260818\_v01.xlsx

The metadata manifest should be mandatory for every artifact:

| Field            | Meaning                                                          |
|------------------|------------------------------------------------------------------|
| artifact_id      | Unique identifier                                                |
| title            | Human-readable artifact name                                     |
| artifact_type    | dissertation / paper / data / model / figure / code / deployment |
| version          | v01, v02, ...                                                    |
| owner            | Akash S. Chatake                                                 |
| supervisor       | Dr. Dipti Jadhav                                                 |
| created_date     | ISO date                                                         |
| related_stage    | outline / mid-sem / final / journal                              |
| source_data      | dataset identifiers used                                         |
| git_commit       | code state                                                       |
| environment_hash | environment or lockfile hash                                     |
| checksum         | SHA256 recommended                                               |
| license          | code/data/license status                                         |
| notes            | brief change description                                         |

The upload order should be:

1. Create GFIS\_2026\_Akash\_Chatake

2. Upload `00_ADMIN` and `01_BITS_ABSTRACT_OUTLINE` first
3. Upload data dictionaries and schemas before models
4. Upload code and reproducibility files before final results
5. Upload dissertation and journal folders only after final figure links are stable
6. Share the final dissertation PDF and presentation directly with your guide

## Priority references and source clusters to cite

The reference strategy should distinguish between **foundational citations** and **application citations**. The dissertation and manuscript should cite the following clusters first.

| Priority cluster                              | Why it matters                                     | Representative source                                                      |
|-----------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------|
| ADM1 original                                 | Establishes mechanistic baseline and complexity    | Batstone et al., ADM1 / IWA model description. <sup>26</sup>               |
| Physics-informed ML foundation                | Justifies integrating physics with ML              | Raissi et al., 2019. <sup>27</sup>                                         |
| Physics-informed ML review                    | Gives broader conceptual framing                   | Karniadakis et al., 2021. <sup>28</sup>                                    |
| LSTM foundation                               | Sequence model legitimacy                          | Hochreiter & Schmidhuber, 1997. <sup>29</sup>                              |
| XGBoost foundation                            | Tabular model legitimacy                           | Chen & Guestrin, 2016. <sup>30</sup>                                       |
| SHAP foundation                               | Explainability method legitimacy                   | Lundberg & Lee, 2017. <sup>22</sup>                                        |
| Calibration foundation                        | ECE / temperature scaling rationale                | Guo et al., 2017. <sup>13</sup>                                            |
| Plant-scale AD ML forecasting                 | Closest modern operational comparison              | Schroer & Just, 2024. <sup>31</sup>                                        |
| Industrial-scale AD biogas prediction         | Applied benchmark for ML in AD                     | Yildirim & Ozkaya, 2023. <sup>32</sup>                                     |
| AutoML / lagged waste effects in co-digestion | Additional applied comparison set                  | Wang et al., 2021. <sup>33</sup>                                           |
| VFA/ALK stability thresholds                  | Stability label rationale                          | Review and operational sources on VFA/alkalinity thresholds. <sup>20</sup> |
| Buswell / BMP theory                          | Methane upper-bound formulation                    | BMP and theoretical methane-potential literature. <sup>19</sup>            |
| Digital twin concept                          | Digital twin framing and wording                   | NASA digital twin sources. <sup>21</sup>                                   |
| Deployment references                         | Justifies implementation stack in methods appendix | FastAPI, Streamlit, SQLite documentation. <sup>34</sup>                    |

## Final positioning

The most defensible version of GFIS is **not** “an AI model for biogas.”

It is:

**a reproducible, physics-guided hybrid AI digital twin for anaerobic digestion that integrates forecasting, soft sensing, explainability, scenario simulation, and safe operating optimization, with a BITS-compliant dissertation path and a Q2-focused publication path.**

That framing is simultaneously aligned with the BITS requirement for **original, non-trivial, 16-week M.Tech-level work** and with the current research direction in **physics-informed scientific machine learning** and **digital decision support for environmental process systems**. 6 35

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- 1 14 26 <https://research.wur.nl/en/publications/the-iwa-anaerobic-digestion-model-no-1-adm1>  
<https://research.wur.nl/en/publications/the-iwa-anaerobic-digestion-model-no-1-adm1>
  - 2 [Sample\\_Project Abstract\\_Outline Report\\_Course No.\\_BITS ID No..docx](https://drive.google.com/file/d/1AY5yticyeFx5SvDQfjiWDOu_9zs9bEcV)  
[https://drive.google.com/file/d/1AY5yticyeFx5SvDQfjiWDOu\\_9zs9bEcV](https://drive.google.com/file/d/1AY5yticyeFx5SvDQfjiWDOu_9zs9bEcV)
  - 8 [Guidelines for preparation of WILP Project Report - 06.04.2022.pdf](https://drive.google.com/file/d/1IwuDXTwCnsUJJ86A8VBNcvyM1VHYvJQ3)  
<https://drive.google.com/file/d/1IwuDXTwCnsUJJ86A8VBNcvyM1VHYvJQ3>
  - 4 [Checklist of Items for the Final Dissertation, Project, Project Work Report.docx](https://drive.google.com/file/d/1l105bLiguTieMXAO6TsCBUah0hBCoGG0)  
<https://drive.google.com/file/d/1l105bLiguTieMXAO6TsCBUah0hBCoGG0>
  - 5 [Steps of Elearn Report Submission..pdf](https://drive.google.com/file/d/1DCrtgBLlrHbWfMAiSsCsbtJRmbK578YR)  
<https://drive.google.com/file/d/1DCrtgBLlrHbWfMAiSsCsbtJRmbK578YR>
  - 6 [NSP 4 Dissertation Orientation PPT S2-25 - 25th Jan 2026.pdf](https://drive.google.com/file/d/13LX_w_H2Q2L85UijVCwWxsMw2GaVZGQw)  
[https://drive.google.com/file/d/13LX\\_w\\_H2Q2L85UijVCwWxsMw2GaVZGQw](https://drive.google.com/file/d/13LX_w_H2Q2L85UijVCwWxsMw2GaVZGQw)
  - 7 16 <https://iro.uiowa.edu/esploro/outputs/dataset/Laboratory-operational-and-SCADA-datasets-for/9984462455902771>  
<https://iro.uiowa.edu/esploro/outputs/dataset/Laboratory-operational-and-SCADA-datasets-for/9984462455902771>
  - 9 <https://www.scimagojr.com/journalsearch.php?clean=0&q=21100933233&tip=sid>  
<https://www.scimagojr.com/journalsearch.php?clean=0&q=21100933233&tip=sid>
  - 10 <https://www.scimagojr.com/journalsearch.php?q=12700154702&tip=sid>  
<https://www.scimagojr.com/journalsearch.php?q=12700154702&tip=sid>
  - 11 <https://www.scimagojr.com/journalsearch.php?clean=0&q=21101072019&tip=sid>  
<https://www.scimagojr.com/journalsearch.php?clean=0&q=21101072019&tip=sid>
  - 12 <https://www.scimagojr.com/journalsearch.php/press.jhu.edu/journalsearch.php?clean=0&q=21101207446&tip=sid>  
<https://www.scimagojr.com/journalsearch.php/press.jhu.edu/journalsearch.php?clean=0&q=21101207446&tip=sid>
  - 13 <https://proceedings.mlr.press/v70/guo17a>  
<https://proceedings.mlr.press/v70/guo17a>

- 15 27 <https://www.sciencedirect.com/science/article/pii/S0021999118307125>  
<https://www.sciencedirect.com/science/article/pii/S0021999118307125>
- 17 31 <https://pubs.acs.org/doi/10.1021/acsestengg.3c00435>  
<https://pubs.acs.org/doi/10.1021/acsestengg.3c00435>
- 18 29 <https://direct.mit.edu/neco/article/9/8/1735/6109/Long-Short-Term-Memory>  
<https://direct.mit.edu/neco/article/9/8/1735/6109/Long-Short-Term-Memory>
- 19 <https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2020.00063/full>  
<https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2020.00063/full>
- 20 <https://www.sciencedirect.com/science/article/pii/S0959152410000272>  
<https://www.sciencedirect.com/science/article/pii/S0959152410000272>
- 21 <https://science.nasa.gov/biological-physical/why-does-the-world-and-nasa-need-digital-twins/>  
<https://science.nasa.gov/biological-physical/why-does-the-world-and-nasa-need-digital-twins/>
- 22 <https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf%7D>  
<https://papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf%7D>
- 23 34 <https://fastapi.tiangolo.com/features/>  
<https://fastapi.tiangolo.com/features/>
- 24 <https://docs.streamlit.io/>  
<https://docs.streamlit.io/>
- 25 <https://www.sqlite.org/docs.html>  
<https://www.sqlite.org/docs.html>
- 28 35 <https://www.nature.com/articles/s42254-021-00314-5>  
<https://www.nature.com/articles/s42254-021-00314-5>
- 30 <https://www.kdd.org/kdd2016/subtopic/view/xgboost-a-scalable-tree-boosting-system>  
<https://www.kdd.org/kdd2016/subtopic/view/xgboost-a-scalable-tree-boosting-system>
- 32 <https://www.sciencedirect.com/science/article/pii/S0045653523012432>  
<https://www.sciencedirect.com/science/article/pii/S0045653523012432>
- 33 <https://seta.lbl.gov/publications/tree-based-automated-machine-learning>  
<https://seta.lbl.gov/publications/tree-based-automated-machine-learning>